

Frappe Operator on OpenShift: Full Deployment Manual

This guide provides the authoritative, step-by-step workflow for deploying the Frappe Operator and ERPNext in an enterprise OpenShift 4.x environment.

1. Prerequisites

- Access to an OpenShift 4.x cluster (v4.10+ recommended).
- `oc` CLI authenticated (`oc login`).
- `helm` CLI (v3+) installed.
- Cluster-admin permissions or equivalent RBAC to create Namespaces, CRDs, and cluster-wide RBAC.

2. Prepare the Infrastructure Project

All operator components and infrastructure (MariaDB, Redis) should reside in a dedicated namespace to maintain clean security boundaries.

```
oc new-project frappe-operator-system
```

3. Install MariaDB Operator (Mandatory)

The Frappe Operator delegates all database lifecycle operations to the [MariaDB Operator](#). This must be installed first.

```
# Add the MariaDB Operator repository
helm repo add mariadb-operator https://mariadb-operator.github.io/mariadb-operator
helm repo update

# Install the MariaDB Operator
helm upgrade --install mariadb-operator mariadb-operator/mariadb-operator \
  --namespace frappe-operator-system \
  --set crds.enabled=true \
  --create-namespace \
  --wait
```

4. Install Frappe Operator

Install the Frappe Operator using the official Helm chart.

```
# Add the Frappe Operator repository
```

```
helm repo add frappe-operator https://vyogotech.github.io/frappe-operator/helm-repo
helm repo update

# Install the operator (disabling nested dependencies as they were manually installed)
helm upgrade --install frappe-operator frappe-operator/frappe-operator \
  --namespace frappe-operator-system \
  --set mariadb-operator.enabled=false \
  --set keda.enabled=false \
  --wait
```

5. Security Note: SCC Compatibility

The Frappe Operator is natively compatible with OpenShift's `restricted-v2` Security Context Constraint (SCC).

- **Dynamic UIDs:** The operator uses `nil` defaults for `runAsUser`, allowing OpenShift to automatically assign a compliant UID.
- **Filesystem Access:** It uses platform-aware volume permissions for GID 0 compatibility.

> [!TIP]

> For deep technical details on SCC handling, see the [OpenShift Technical Guide](#).

6. Preparing OpenShift-Compatible Container Images

OpenShift requires containers to run as non-root (GID 0). The official images must be patched. This is typically a one-time activity per ERP release.

6.1 Base Image – OpenShift Compatibility Patch

Use this container file to create a base image.

```
# ?? OpenShift-Compatible ERPNext Base Image ??????????????????????????
FROM frappe/erpnext:v15.100.0 AS erpnext

USER root

# 1. Nginx and Permission Fixes for GID 0
RUN rm -fr /etc/nginx/sites-enabled/default && \
  sed -i '/user www-data/d' /etc/nginx/nginx.conf && \
  sed -i 's/listen 80 default_server/listen 8080 default_server/' /etc/nginx/sites-available/default
|| true && \
  ln -sf /dev/stdout /var/log/nginx/access.log && ln -sf /dev/stderr /var/log/nginx/error.log && \
  touch /run/nginx.pid && \
  chown -R frappe:root /etc/nginx /var/log/nginx /var/lib/nginx /run && \
  chmod -R g=u /etc/nginx /var/log/nginx /var/lib/nginx /run

# 2. Copy Helper Scripts
COPY resources/nginx-template.conf /templates/nginx/frappe.conf.template
```

Build and Push Base Image

6.2 Custom-Apps Image – Two-Stage Build

1. Define apps.json

2. Create the Custom Containerfile

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3. Build and Push Custom Image

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7. Deployment Examples

With the infrastructure ready and images built, you can now deploy your Bench and Site.

Step 7.1: Create MariaDB Shared Instance

Create `mariadb-instance.yaml`:

```
apiVersion: v1
kind: Secret
metadata:
  name: frappe-mariadb-root
  namespace: frappe-operator-system
type: Opaque
stringData:
  password: "StrongRootPassword123"
```

```
apiVersion: k8s.mariadb.com/v1alpha1
kind: MariaDB
metadata:
  name: frappe-mariadb
  namespace: frappe-operator-system
spec:
  rootPasswordSecretKeyRef:
    name: frappe-mariadb-root
    key: password
  image: mariadb:10.11
  storage:
    size: 20Gi
  resources:
    requests:
      cpu: 250m
      memory: 512Mi
  replicas: 1
```

Step 7.2: Create the FrappeBench

A Bench represents your application environment. Create `my-bench.yaml`. **This uses the custom image built in Section 6.2.**

```
apiVersion: vyogo.tech/v1alpha1
kind: FrappeBench
metadata:
  name: prod-bench
  namespace: frappe
spec:
  frappeVersion: "15.0.0"
  imageConfig:
    repository: your-registry.io/frappe/custom-apps
    tag: v15-1.0.0
    pullPolicy: Always
    pullSecrets:
      - name: registry-credentials
```

```
componentAutoscaling:
  unicorn:
    enabled: true
    minReplicas: 2
    maxReplicas: 5
    provider: hpa
    hpa:
      metric: cpu
      targetUtilization: 75
      scaleUpStabilization: 0
      scaleDownStabilization: 60
  nginx:
    enabled: true
    minReplicas: 2
    maxReplicas: 4
    provider: hpa
    hpa:
      metric: cpu
      targetUtilization: 80
      scaleUpStabilization: 0
      scaleDownStabilization: 60
  workerDefault:
    enabled: true
    minReplicas: 1
    maxReplicas: 5
    provider: hpa
    hpa:
      metric: cpu
      targetUtilization: 85

  workerShort:
    enabled: false # Static replicas
    staticReplicas: 1

  workerLong:
    enabled: false
    staticReplicas: 1

  socketio:
    enabled: false
    staticReplicas: 1

storageSize: 5Gi
```

Step 7.3: Create the FrappeSite

A Site is your actual application instance. Create `my-site.yaml`:

```
apiVersion: v1
kind: Secret
metadata:
  name: prod-site-admin
  namespace: frappe
```

```
type: Opaque
stringData:
  password: "AdminPassword123"
```

```
apiVersion: vyogo.tech/v1alpha1
kind: FrappeSite
metadata:
  name: prod-site
  namespace: frappe
spec:
  benchRef: prod-bench # point to the bench created above.
  siteName: prod-site.apps.cluster.example.com # Must be a valid domain
  adminPasswordSecretRef:
    name: prod-site-admin
    key: password
  dbConfig:
    mode: shared
    mariadbRef:
      name: frappe-mariadb
  routeConfig:
    enabled: true
    termination: edge
```

Apply all manifests:

```
oc apply -f mariadb-instance.yaml
oc apply -f my-bench.yaml
oc apply -f my-site.yaml
```

8. Verification & Troubleshooting

Check Pods and Jobs

```
oc get pods -n frappe-operator-system
oc get jobs -n frappe-operator-system
```

Access the Site

Once the site phase is `Ready`, retrieve the URL from the OpenShift Route:

```
oc get route prod-site -n frappe-operator-system -o jsonpath='{.spec.host}'
```

For more details, see:

- [OpenShift Technical Guide](#)
- [MariaDB Integration Guide](#)